

MuseCritic: Learning Multi-Aspect Song Rewards through Natural-Language Aesthetic Critiques

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Long-form song generation models continue to improve in duration, structural integrity, and acoustic complexity, making reliable aesthetic rewards increasingly important for aligning these models with human preferences. However, reward models for complete songs remain limited, and existing evaluators typically predict scores in a single forward pass without providing readable explanations. To this end, we introduce MuseCritic, a semi-scalar reward model that generates a natural-language critique covering five aesthetic dimensions and uses it as an intermediate representation to predict continuous reward scores. MuseCritic follows a two-stage training pipeline: a teacher model first provides high-quality critiques for supervised fine-tuning, after which the fine-tuned model generates its own critiques for reward learning, mitigating distribution shift between training and inference. On an in-domain test set of 200 SongEval songs, MuseCritic reduces macro-averaged mean squared error from 0.2875 to 0.2316 and improves macro-averaged LCC, SRCC, and Kendall’s τ to 0.9068, 0.8838, and 0.7178, respectively. On the out-of-domain Music Arena benchmark with 733 preference pairs, it achieves the highest accuracy of 71.35%. Moreover, using MuseCritic with GRPO improves Muse-0.6B on all nine aesthetic metrics from SongEval and Audiobox Aesthetics. These results demonstrate that critique-conditioned reward modeling reduces scoring error and provides an effective optimization signal for song generation. The project repository is available at <https://github.com/WuqnEl/MuseCritic>.

1. Introduction

Text-to-music generation has progressed from synthesizing short clips to composing complete songs. Early systems generate musical excerpts from natural-language descriptions (Agostinelli et al., 2023; Copet et al., 2023), while recent systems—including DiffRhythm, YuE, LeVo, ACE-Step, and Muse—support minutes-long vocal music, lyric alignment, and fine-grained style control (Gong et al., 2025; Jiang et al., 2026; Lei et al., 2025; Ning et al., 2025; Yuan et al., 2025). As generation length and acoustic realism improve, reliable aesthetic evaluation becomes increasingly important for model selection and for providing optimization signals aligned with human preferences.

Existing evaluators, however, remain limited for complete songs. Objective metrics such as phoneme error rate, MuLan, and MuQ-MuLan capture lyric intelligibility or audio–text correspondence rather than perceptual and artistic quality (Huang et al., 2022; Zhu et al., 2025). PAM, MusicEval, and Audiobox Aesthetics broaden evaluation to audio quality and musical aesthetics (Deshmukh

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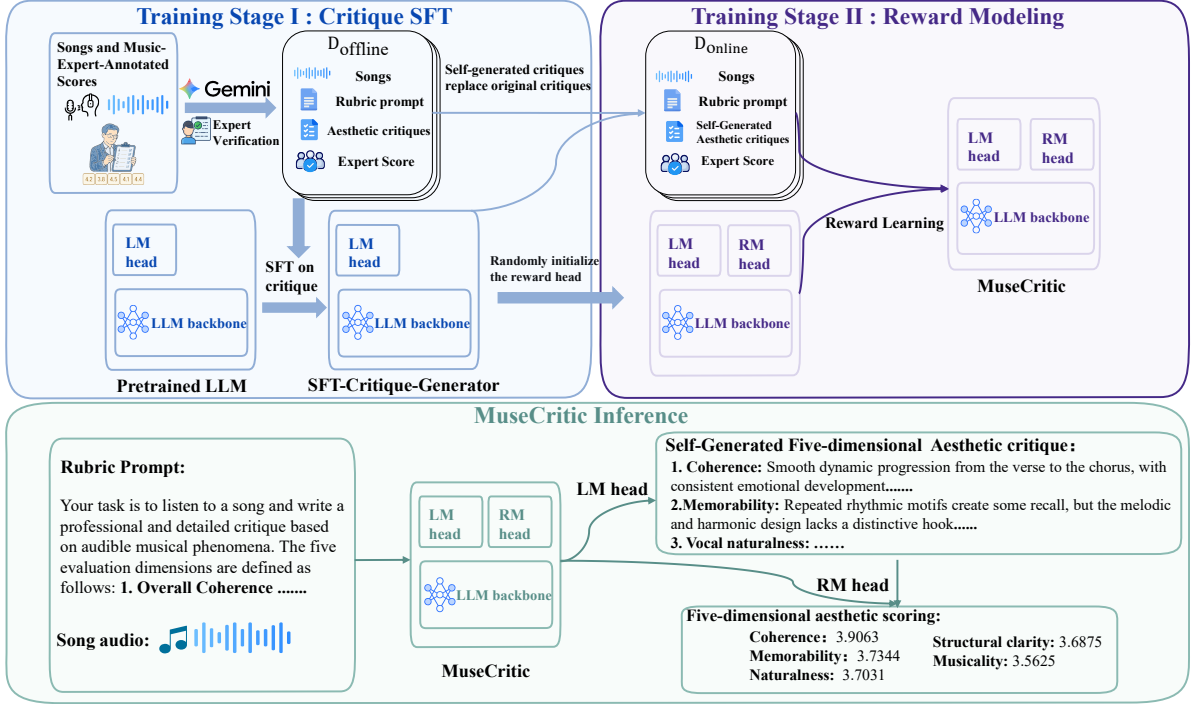


Figure 1 | Overview of the two-stage training and inference pipeline of MuseCritic. In Stage I, Gemini-3-Pro generates aesthetic critiques from songs, evaluation rubrics, and expert mean ratings to construct $\mathcal{D}_{\text{off}}$, which is used to fine-tune a pretrained large language model into an SFT critique generator. In Stage II, the SFT model replaces the teacher critiques with one self-generated five-aspect critique per song to construct $\mathcal{D}_{\text{on}}$; a scalar reward head is then initialized, and MuseCritic is trained using expert mean ratings as regression targets. At inference time, MuseCritic first generates a five-aspect critique and then predicts five continuous aesthetic scores conditioned on the song, rubric, and self-generated critique.

et al., 2024; Liu et al., 2025a; Tjandra et al., 2025), while SongEval provides five expert aesthetic criteria for full-length synthetic songs (Yao et al., 2025); nevertheless, these evaluators map audio directly to scores and do not explain the audible evidence behind their judgments. In contrast, LLM-as-a-Judge and critique-conditioned reward models show that language models can interpret rubrics and generate corresponding rationales or supporting reasoning before scoring textual responses, thereby improving textual preference modeling (Ankner et al., 2024; Liu et al., 2025b; Yu et al., 2025; Zhang et al., 2024; Zheng et al., 2023). Whether this linguistic intermediate representation can improve multidimensional aesthetic reward modeling for complete songs remains underexplored. Existing audio reward models predict scores directly, while general-purpose multimodal models lack task-specific training and show weaker agreement with expert judgments.

In this work, we investigate explicit linguistic reasoning in song reward modeling, enabling the model to form a representation of aesthetic evidence aligned with the evaluation rubric before predicting multidimensional scores. To this end, we propose MuseCritic, a semi-scalar reward model built around natural-language aesthetic critiques. Given a song and a five-dimensional evaluation rubric, MuseCritic generates a single critique containing five labeled analyses: Overall Coherence, Memorability, Naturalness of Vocal Breathing and Phrasing, Clarity of Song Structure, and Overall Musicality. Conditioned on the song, rubric, and self-generated critique, a reward head then predicts five continuous aesthetic scores ranging from 1 to 5. The critique is not a post-hoc text appended to predicted scores; it is an explicit intermediate variable connecting song representations to rewards.

This design introduces a variable-length textual representation before prediction and organizes auditory evidence distributed across a long song into dimension-aligned descriptions.

Figure 1 summarizes the training and inference pipeline. The model first learns five-dimensional aesthetic analysis from critiques generated by an external teacher. It then generates self-generated critiques for the training songs and performs reward learning on the resulting dataset.

Across both in-domain and out-of-domain evaluations, critique-conditioned reward modeling consistently improves agreement with human judgments. On the held-out SongEval test set, MuseCritic reduces macro-averaged MSE from 0.2875 to 0.2316 and raises macro-averaged LCC, SRCC, and Kendall’s τ to 0.9068, 0.8838, and 0.7178. On Music Arena, MuseCritic attains the highest observed pairwise accuracy of 71.35% across 733 preference pairs. When used as the reward model for GRPO, it further improves Muse-0.6B on all nine aesthetic metrics reported by SongEval and Audiobox Aesthetics.

Our contributions are threefold:

- We introduce MuseCritic, a semi-scalar reward model for complete songs. It generates a natural-language critique containing five dimension-specific analyses and uses this critique to predict five continuous scores, combining verbal evaluation and reward estimation within a single model (Figure 1).
- We systematically evaluate MuseCritic on in-domain SongEval ratings and out-of-domain Music Arena preferences. Self-generated critiques improve agreement with expert scores and mitigate the critique distribution shift between reward training and inference (Tables 1, 2, and 4).
- We use MuseCritic as the reward model for GRPO training of Muse-0.6B. The resulting policy improves on all nine observed metrics from SongEval and Audiobox Aesthetics, showing that the learned reward provides a useful optimization signal (Table 3).

2. Related Work

2.1. Text-to-Song Generation

Advances in neural audio codecs and large-scale audio generation have moved language-conditioned music synthesis from early systems such as MusicLM and MusicGen toward recent systems capable of generating complete songs (Agostinelli et al., 2023; Copet et al., 2023). Unlike short musical excerpts, complete songs require cross-section structure, coordination between vocals and accompaniment, and long-range lyric alignment.

Recent systems can generate complete songs several minutes in duration. DiffRhythm performs end-to-end long-form song synthesis through latent diffusion (Ning et al., 2025); YuE combines dual-track tokenization with progressive structural conditioning to generate songs up to five minutes long (Yuan et al., 2025); and ACE-Step integrates diffusion with musical semantic representations (Gong et al., 2025). LeVo and Muse further support vocal-accompaniment coordination and fine-grained, multi-turn style control (Jiang et al., 2026; Lei et al., 2025). MusicRL learns music rewards from human feedback (Cideron et al., 2024), while LeVo uses multi-preference DPO to improve song quality and instruction following (Lei et al., 2025). As complete-song generation matures, reliable aesthetic evaluation and reward signals suitable for reinforcement learning become central challenges. Our work focuses on the reward itself, enabling one model to both describe five-dimensional aesthetic quality and assign continuous rewards.

2.2. Reward Modeling and Music Aesthetic Evaluation

Classical RLHF reward models attach a shallow scalar head to a pretrained language-model backbone to fit human preferences (Ouyang et al., 2022), whereas LLM-as-a-Judge uses capable language models to produce evaluations and supporting rationales (Zheng et al., 2023). Semi-scalar and generative reward models seek to combine linguistic explanation with numerical prediction. CLOUD generates natural-language critiques before predicting rewards (Ankner et al., 2024); Critic-RM jointly trains self-generated critiques and reward prediction (Yu et al., 2025); and Generative Verifiers and DeepSeek-GRM improve reward judgments through reasoning traces and test-time sampling (Liu et al., 2025b; Zhang et al., 2024). These approaches primarily evaluate text responses. MuseCritic extends this modeling paradigm to long-form song audio and multidimensional continuous aesthetic assessment.

Existing music evaluators typically predict proxy metrics or scalar scores. MuLan and MuQ-MuLan measure text–music correspondence (Huang et al., 2022; Zhu et al., 2025); PAM estimates reference-free audio quality (Deshmukh et al., 2024); MusicEval trains a CLAP-based evaluator on expert ratings of generated music clips (Liu et al., 2025a); and Audiobox Aesthetics predicts content enjoyment, content usefulness, production complexity, and production quality (Tjandra et al., 2025). SongEval defines five dimensions for complete Chinese and English songs: coherence, memorability, vocal naturalness, structural clarity, and musicality (Yao et al., 2025), but still targets score prediction alone. MuseCritic adopts the SongEval rubric while retaining the critique-generation capability of an audio language model, allowing explicit aesthetic evidence to participate in reward prediction.

3. Method

3.1. Preliminaries

Problem formulation. Let $x \in \mathcal{X}$ denote a complete song with vocals, p an aesthetic-evaluation prompt that defines the assessment dimensions, and $K = 5$ the number of dimensions. The expert annotation is $\mathbf{y}^* = [y_1^*, \dots, y_K^*] \in [1, 5]^K$. The five dimensions are Overall Coherence, Memorability, Naturalness of Vocal Breathing and Phrasing, Clarity of Song Structure, and Overall Musicality (Yao et al., 2025). We use an instruction-tuned audio language model f_θ as the shared backbone of our semi-scalar reward model. Its critique generator g_c and aesthetic scorer g_s are

$$g_c = h_{\text{LM}} \circ f_\theta, \quad g_s = h_{\text{RM}} \circ f_\theta, \quad (1)$$

where h_{LM} is a language-modeling head that generates a natural-language aesthetic critique, and h_{RM} is a reward-modeling head that produces K continuous aesthetic scores. The two components always share the same backbone parameters θ .

Reward modeling. Unlike reward models that predict a scalar directly from audio, the aesthetic scorer of MuseCritic jointly conditions on the song x , evaluation rubric p , and aesthetic critique c :

$$\hat{\mathbf{y}} = g_s(x, p, c). \quad (2)$$

For any set $\mathcal{S}$ of expert-annotated samples, we learn the five-dimensional scores by minimizing mean squared error:

$$\mathcal{L}_{\text{MSE}}(\mathcal{S}; g_s) = \frac{1}{|\mathcal{S}|K} \sum_{(x, p, c, \mathbf{y}^*) \in \mathcal{S}} \|g_s(x, p, c) - \mathbf{y}^*\|_2^2. \quad (3)$$

This objective simultaneously constrains all five dimensions to align the predictions with expert ratings.

3.2. The MuseCritic Reward Model

Overview. To introduce an explicit textual intermediate representation into reward modeling, we treat natural-language aesthetic critiques as intermediate variables between a song and its final scores. Given a song x and rubric p , MuseCritic first generates

$$\hat{c} = g_c(x, p), \quad (4)$$

where $\hat{c}$ is a single critique with five labeled sections describing perceptible musical phenomena under the corresponding dimensions. The shared backbone then receives the song, rubric, and self-generated critique to produce

$$\hat{y} = g_s(x, p, \hat{c}). \quad (5)$$

Relative to direct score prediction, Equation 5 introduces a variable-length textual representation before scoring and organizes descriptions of auditory evidence distributed across a long song according to the evaluation rubric.

Training MuseCritic. Ideally, MuseCritic would be trained on a dataset $\mathcal{D}^* = \{(x_i, p, c_i^*, y_i^*)\}_{i=1}^N$, where c_i^* is a human-authored aesthetic critique consistent with the corresponding expert ratings. Such critiques would be authored by human music experts, but existing song datasets rarely pair expert scores with expert critiques. Following prior work that constructs supervision from model feedback (Ankner et al., 2024; Bai et al., 2022), we use the audio-capable Gemini-3-Pro (Google, 2025) to generate score-conditioned synthetic critiques. Specifically, we begin with a score-only dataset $\mathcal{D}_{\text{score}} = \{(x_i, y_i^*)\}_{i=1}^N$. To generate a critique, Gemini observes the song x_i , rubric p , and expert mean ratings y_i^* , then writes dimension-specific commentary whose evaluative polarity is consistent with each rating on a five-point Likert scale. The prompt requests analysis of concrete musical phenomena while prohibiting numerical scores in the critique itself. This procedure yields the offline critique dataset

$$\mathcal{D}_{\text{off}} = \{(x_i, p, c_i^{\text{off}}, y_i^*)\}_{i=1}^N. \quad (6)$$

Appendix D provides the complete external-teacher prompt. To further ensure the reliability of the aesthetic critiques, we manually verify the generated critiques, as detailed in Appendix A.

Training comprises two stages. In Stage I, we supervise the audio language model on $\mathcal{D}_{\text{off}}$ with the critique-generation objective

$$\mathcal{L}_{\text{SFT}} = - \frac{1}{\sum_{i=1}^N |c_i^{\text{off}}|} \sum_{i=1}^N \sum_{t=1}^{|c_i^{\text{off}}|} \log p_{\theta}(c_{i,t}^{\text{off}} \mid x_i, p, c_{i,<t}^{\text{off}}). \quad (7)$$

This is the standard next-token cross-entropy averaged over non-masked critique target tokens; the audio and user-prompt tokens do not contribute to the loss. This stage produces a critique generator g'_c with the ability to analyze a song along the five aesthetic dimensions.

In Stage II, we first use g'_c to generate a new critique for every training example:

$$c_i^{\text{on}} = g'_c(x_i, p). \quad (8)$$

We replace the external-teacher critiques with these self-generated critiques and call the resulting collection the self-generated critique dataset $\mathcal{D}_{\text{on}}$:

$$\mathcal{D}_{\text{on}} = \{(x_i, p, c_i^{\text{on}}, y_i^*)\}_{i=1}^N. \quad (9)$$

We then initialize the shared MuseCritic backbone from the SFT model and jointly update the backbone LoRA parameters and all reward-head parameters on $\mathcal{D}_{\text{on}}$. The reward model thus learns from critiques generated by the target model family rather than by the external teacher, mitigating the critique distribution shift between reward training and inference. The objective is the MSE from Equation 3 applied to $\mathcal{D}_{\text{on}}$:

$$\mathcal{L}_{\text{MC}} = \mathcal{L}_{\text{MSE}}(\mathcal{D}_{\text{on}}; g_s). \quad (10)$$

At inference time, the final LoRA-updated MuseCritic backbone generates an aesthetic critique according to Equation 4 and then predicts five rewards according to Equation 5.

4. Experiments

4.1. Experimental Setup

Data. We use SongEval (Yao et al., 2025) for training and in-domain evaluation. It contains 2,399 complete Chinese and English songs totaling more than 140 hours of audio, with five-dimensional aesthetic scores from 16 annotators with music expertise. The data span nine major genres, including pop, rock, and electronic music. Because the original work does not release a fixed train–test split, we shuffle all samples with seed 42, use 2,199 songs for training, and reserve the remaining 200 for testing. Every experiment uses this split. For the training set, Gemini-3-Pro (Google, 2025) observes each song and its expert mean ratings to generate a single critique containing five dimension-specific analyses, producing $\mathcal{D}_{\text{off}}$. The SFT model subsequently regenerates one five-part critique per song with greedy decoding ($T = 0$, no sampling), producing the self-generated critique dataset $\mathcal{D}_{\text{on}}$. Appendices D and A provide the teacher and model prompts and the splitting algorithm.

For out-of-domain evaluation, we use the 733 song-preference pairs from Music Arena (Kim et al., 2025). This dataset contains outputs from systems not represented in SongEval and provides pairwise human preferences rather than five-dimensional absolute scores. It therefore measures preference generalization under distribution shift.

Baselines. The in-domain task requires five-dimensional song aesthetic scores. We compare against two classes of baselines: (1) **Gemini-3.1-Pro** (Google DeepMind, 2026), an audio-capable LLM-as-a-Judge that generates critiques and five-dimensional ratings under the same rubric; and (2) **SongEval (UTMOS)**, a discriminative reward model that directly regresses five scores from audio. SongEval does not release its original training code or data split, so we reimplement its published method and UTMOS architecture and train it for ten epochs on our 2,199/200 split, controlling for differences in training data.

Music Arena requires pairwise preference prediction. In addition to MuseCritic, we evaluate the officially released SongEval weights (Yao et al., 2025), the general-purpose audio aesthetic model Audiobox Aesthetics (Tjandra et al., 2025), and the open-source audio language model Qwen3-Omni-30B-A3B-Instruct (Xu et al., 2025). Qwen3-Omni uses the same rubric and generative scoring format as the in-domain Gemini baseline.

Metrics. For in-domain evaluation, we report dimension-wise mean squared error (MSE), Pearson’s linear correlation coefficient (LCC), Spearman’s rank correlation coefficient (SRCC), and Kendall’s τ (KTAU). MSE measures agreement with the experts’ absolute rating scale, while the three correlation coefficients measure relative agreement across songs in terms of linear association and ranking. Each metric is computed independently over the 200 test songs for each dimension.

For out-of-domain evaluation, we report pairwise accuracy. A prediction is correct only when the

Dimension	Model	MSE ↓	LCC ↑	SRCC ↑	KTAU ↑
Coherence	MuseCritic	0.2058	0.9147	0.8894	0.7262
	Gemini-3.1-Pro	0.9881	0.5906	0.5898	0.4359
	SongEval (UTMOS)	0.2605	0.8876	0.8605	0.6835
Memorability	MuseCritic	0.2511	0.9098	0.8839	0.7232
	Gemini-3.1-Pro	1.1183	0.5806	0.5623	0.4113
	SongEval (UTMOS)	0.3055	0.8857	0.8595	0.6837
Vocal naturalness	MuseCritic	0.2537	0.8906	0.8678	0.6945
	Gemini-3.1-Pro	1.9736	0.4921	0.5237	0.3877
	SongEval (UTMOS)	0.3071	0.8593	0.8293	0.6491
Structural clarity	MuseCritic	0.2275	0.9068	0.8806	0.7109
	Gemini-3.1-Pro	1.4740	0.5532	0.5752	0.4162
	SongEval (UTMOS)	0.2730	0.8831	0.8565	0.6790
Musicality	MuseCritic	0.2202	0.9120	0.8972	0.7344
	Gemini-3.1-Pro	0.9763	0.6152	0.6099	0.4459
	SongEval (UTMOS)	0.2913	0.8807	0.8599	0.6876

Table 1 | In-domain five-dimensional aesthetic scoring on the SongEval test set. All models are evaluated on the fixed test set of 200 songs. SongEval (UTMOS) denotes the regressor retrained on the same 2,199/200 split. MSE measures absolute error against expert ratings; LCC, SRCC, and KTAU denote Pearson, Spearman, and Kendall τ correlations. Arrows indicate the preferred direction, and bold marks the best value for each metric and dimension.

model assigns a strictly higher overall reward to the human-preferred song; ties count as errors. For SongEval, Audiobox Aesthetics, and MuseCritic, we average the respective output dimensions with equal weight. For Qwen3-Omni, we parse its five generated ratings under the shared rubric and take their mean. All main evaluations use deterministic decoding with $T = 0$.

For downstream evaluation after GRPO, music quality is assessed using SongEval (Yao et al., 2025) in terms of overall coherence (CO), memorability (ME), naturalness of vocal breathing and phrasing (NA), clarity of song structure (CL), and overall musicality (MU). Audio quality is evaluated with Audiobox Aesthetics (Tjandra et al., 2025), which considers content enjoyment (CE), content usefulness (CU), production complexity (PC), and production quality (PQ).

Training. All MuseCritic variants use MOSS-Audio-8B-Instruct (Yang et al., 2026) as the base model. During critique cold-start, we fine-tune all backbone parameters with a learning rate of 5×10^{-5} and use the checkpoint saved after one epoch. We initialize MuseCritic from this checkpoint and train it for ten epochs on the self-generated critique data. The backbone uses LoRA (rank 8, scaling factor 32) (Hu et al., 2021), while the reward head is fully updated with a learning rate of 2×10^{-4} . Both optimization stages use four NVIDIA H200 GPUs, a per-device batch size of 1, gradient accumulation over 8 steps (global batch size 32), a 5% warmup, cosine learning-rate scheduling, and a fixed random seed of 42. Reward learning uses a weight decay of 0.1. The offline-critique and no-critique ablations use exactly the same reward-learning configuration. Appendix C lists all hyperparameters.

4.2. Evaluation Results

4.2.1. In-Domain Song Aesthetic Scoring

Table 1 compares model predictions with the expert mean ratings of the 200-song held-out SongEval test set. MuseCritic achieves the lowest MSE and highest correlation on all five dimensions. Relative to the SongEval (UTMOS) regression baseline trained on the same data, MuseCritic reduces macro-

Model	Accuracy (%) $\uparrow$
Qwen3-Omni-30B-A3B	53.75
Audiobox Aesthetics	68.49
SongEval	70.80
MuseCritic	71.35

Table 2 | Out-of-domain pairwise preference results on Music Arena. The test set contains 733 song pairs with human preference labels. A prediction is correct when the model assigns a higher mean aesthetic score to the human-preferred song; ties count as errors. SongEval uses its officially released weights. MuseCritic achieves the highest observed accuracy.

averaged MSE from 0.2875 to 0.2316. Macro-averaged LCC, SRCC, and KTAU increase from 0.8793, 0.8531, and 0.6766 to 0.9068, 0.8838, and 0.7178. Section 4.4 isolates the effect of critique modeling through controlled ablations.

Although Gemini-3.1-Pro has not been specialized for song aesthetics, it generates coherent critiques. Its macro-average MSE of 1.3061, however, substantially exceeds the 0.2316 of MuseCritic, and its dimension-wise correlations fall below those of models supervised by expert scores. This result indicates that an unadapted LLM-as-a-Judge cannot reliably reproduce five-dimensional expert judgments of song aesthetics. By supervising the reward head with expert ratings, MuseCritic maps information in the critiques to continuous scores that better align with expert assessments.

4.2.2. Out-of-Domain Preference Generalization

Table 2 reports pairwise preference accuracy on Music Arena. MuseCritic achieves the highest observed accuracy of 71.35%, which is 0.55, 2.86, and 17.60 percentage points above SongEval, Audiobox Aesthetics, and Qwen3-Omni, respectively. Together with the in-domain results, this observation suggests that the learned rewards retain ranking information under an out-of-domain preference evaluation.

4.3. Downstream Reinforcement Learning

To test whether MuseCritic provides a useful optimization signal, we use it to train Muse-0.6B (Jiang et al., 2026) with GRPO (Shao et al., 2024). Using seed 42, we sample 500 multi-turn song-generation examples from the public Muse training set while preserving the original language ratio: 195 Chinese and 305 English examples. We define the reinforcement-learning reward function as the mean of the five aesthetic scores assigned by MuseCritic to each generated song $\tilde{x}$:

$$r(\tilde{x}) = \frac{1}{K} \sum_{k=1}^K \hat{y}_k(\tilde{x}). \quad (11)$$

We train for one epoch on six H200 GPUs, sample eight outputs per prompt, and use a learning rate of 10^{-6} . Appendix F gives the complete rollout configuration.

Evaluation uses the 100 multi-turn test prompts from Muse (Jiang et al., 2026). Both Muse and Muse-GRPO use greedy decoding with $T = 0$. Appendix F describes the complete evaluation protocol and robustness checks across evaluation windows. As shown in Table 3, Muse-GRPO obtains higher observed values than the original model on all nine Audiobox Aesthetics and SongEval metrics. The largest Audiobox gains occur in production complexity (PC, +0.15) and content enjoyment (CE, +0.12), while all five SongEval dimensions improve by 0.05–0.06. The consistent numerical gains

Model	Audiobox Aesthetics $\uparrow$				SongEval $\uparrow$				
	CE	CU	PC	PQ	CO	MU	ME	CL	NA
Muse	7.33	7.64	6.37	8.11	4.01	3.84	3.94	3.89	3.83
MUSE-GRPO	7.45	7.66	6.52	8.12	4.07	3.89	4.00	3.94	3.89

Table 3 | Effect of MuseCritic-guided GRPO on the aesthetic quality of Muse-0.6B. Results use the 100 multi-turn Muse test prompts and $T = 0$ decoding. CE/CU/PC/PQ denote Audiobox content enjoyment, content usefulness, production complexity, and production quality. CO/MU/ME/CL/NA denote SongEval coherence, musicality, memorability, structural clarity, and vocal naturalness. Arrows indicate the preferred direction, and bold marks the better observed result in each column. Muse-GRPO yields numerical gains on all nine metrics.

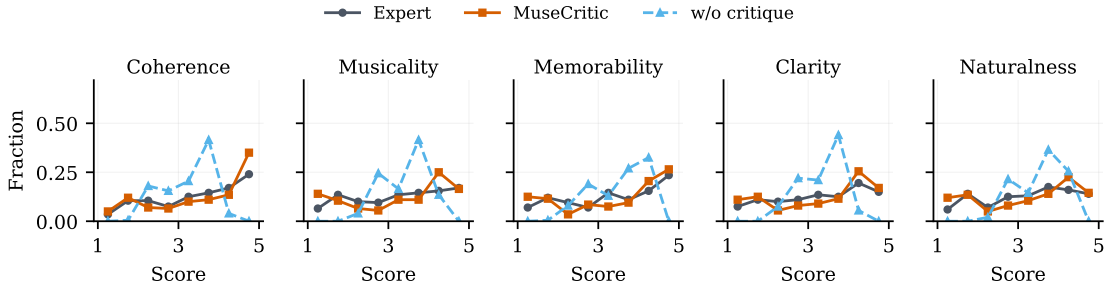


Figure 2 | Effect of natural-language aesthetic critiques on five-dimensional score distributions. Curves show expert ratings, full MuseCritic predictions, and predictions from the no-critique variant on the fixed SongEval test set. Coherence, Musicality, Memorability, Clarity, and Naturalness correspond to the five aesthetic dimensions. Without critiques, predictions cluster in a narrow high-score region; the full model more closely matches the range of expert ratings.

across two song-aesthetic evaluators suggest that the MuseCritic reward captures several aesthetic dimensions.

4.4. Ablation Studies

Table 4 studies three design choices across all five dimensions: natural-language aesthetic critiques, self-generated critiques, and SFT initialization before reward learning. All variants use the same training songs, reward head, and optimization configuration.

Self-generated aesthetic critiques improve absolute-score agreement. Removing the critiques increases macro-average MSE from 0.2316 to 0.5005, while the macro-averaged rank correlations remain close. Figure 2 explains this difference: the no-critique model concentrates most predictions in a narrow range around 4. It can therefore preserve a coarse ranking among songs without recovering the dynamic range of expert ratings. A reward model should preserve not only rank order but also the relative magnitude of quality differences through score intervals. An overly contracted prediction distribution reduces agreement with experts’ absolute ratings and weakens reward discrimination among samples of different quality during downstream policy optimization. With critiques, evaluative polarity and concrete musical descriptions provide an explicit semantic reference, bringing the prediction distribution closer to the expert scale and reducing MSE in every dimension.

Dimension	Model variant	MSE ↓	LCC ↑	SRCC ↑	KTAU ↑
Coherence	MuseCritic	0.2058	0.9147	0.8894	0.7262
	w/o critique	0.3881	0.9173	0.8954	0.7326
	Offline critique	0.4821	0.8342	0.7566	0.5675
	w/o SFT	0.8556	0.6300	0.6066	0.4689
Memorability	MuseCritic	0.2511	0.9098	0.8839	0.7232
	w/o critique	0.4936	0.9048	0.8842	0.7219
	Offline critique	0.6635	0.8141	0.7590	0.5672
	w/o SFT	1.0037	0.6175	0.6378	0.4823
Vocal naturalness	MuseCritic	0.2537	0.8906	0.8678	0.6945
	w/o critique	0.6186	0.8899	0.8684	0.7034
	Offline critique	0.5739	0.7936	0.7527	0.5674
	w/o SFT	0.6469	0.6779	0.6768	0.4978
Structural clarity	MuseCritic	0.2275	0.9068	0.8806	0.7109
	w/o critique	0.4635	0.9041	0.8785	0.7088
	Offline critique	0.5033	0.8145	0.7479	0.5526
	w/o SFT	0.6681	0.6869	0.6764	0.5061
Musicality	MuseCritic	0.2202	0.9120	0.8972	0.7344
	w/o critique	0.5386	0.9080	0.8875	0.7293
	Offline critique	0.5175	0.8205	0.7765	0.5877
	w/o SFT	0.5960	0.7354	0.7325	0.5488

Table 4 | In-domain ablations of the key MuseCritic training choices. All variants use the same 2,199/200 SongEval split. “w/o critique” predicts scores directly, “Offline critique” trains on Gemini critiques, and “w/o SFT” generates training critiques with the unadapted MOSS-Audio model. The full model combines SFT initialization with self-generated critiques. Arrows indicate the preferred direction, and bold marks the best value for each metric and dimension.

Training on self-generated critiques improves reward prediction. To isolate the role of the self-generated dataset $\mathcal{D}_{\text{on}}$, we train a control model on $\mathcal{D}_{\text{off}}$, using external-teacher critiques instead of critiques generated by the target model. As Table 4 shows, the full MuseCritic trained on $\mathcal{D}_{\text{on}}$ achieves higher LCC, SRCC, and KTAU than the offline-critique variant on every dimension. Replacing $\mathcal{D}_{\text{on}}$ with $\mathcal{D}_{\text{off}}$ increases macro-average MSE from 0.2316 to 0.5481 and decreases macro-average LCC, SRCC, and KTAU from 0.9068, 0.8838, and 0.7178 to 0.8154, 0.7585, and 0.5685, respectively. We suggest that the external teacher and MOSS-Audio differ in vocabulary, information density, and patterns. Training only on teacher critiques therefore creates a larger mismatch between the critiques observed during reward training and those generated at inference time.

SFT initializes effective critique generation. Without SFT, we use the unadapted MOSS-Audio-8B-Instruct model to generate critiques and subsequently train the reward model on these critiques. This variant performs worst overall: macro-averaged MSE increases from 0.2316 to 0.7541, while macro-averaged LCC, SRCC, and KTAU decrease from 0.9068, 0.8838, and 0.7178 to 0.6695, 0.6660, and 0.5008, respectively. The degradation in both absolute-score accuracy and ranking consistency indicates that self-generation alone does not produce a reliable intermediate representation for reward learning. Supervised fine-tuning on teacher critiques is therefore necessary for the critique generator to acquire the evaluation criteria, domain-specific terminology, and rating semantics required to distinguish both the level and relative ordering of song quality.

5. Conclusion

We introduced MuseCritic, a semi-scalar reward model for the aesthetic evaluation of complete songs. The model treats natural-language aesthetic critiques as intermediate variables: it first organizes

evidence along five dimensions and then predicts continuous reward scores. Training initializes critique generation with external-teacher supervision and trains the reward head on self-generated critiques to mitigate the critique distribution shift between training and inference. Experiments on SongEval, Music Arena, and three controlled ablations show that SFT-initialized, self-generated critiques reduce absolute scoring error while maintaining strong rank correlations. Using MuseCritic to train Muse with GRPO improves all nine metrics from two song-aesthetic evaluators, connecting critique-conditioned evaluation to preference optimization.

Limitations

The critique-then-score procedure requires autoregressive critique generation before reward prediction and is therefore more computationally expensive than direct score regression. Moreover, long-form song datasets that pair audio with expert aesthetic ratings remain scarce. Consequently, MuseCritic is trained primarily on the Chinese and English vocal songs in SongEval. Future work can extend training to a broader range of languages and musical genres.

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A. Data Construction and Splitting

SongEval split. We load all 2,399 SongEval examples and shuffle them once using random seed 42. The first 2,199 examples form the training set, and the final 200 examples form the test set. We construct the split before generating critiques and reuse the same test examples across all main experiments, baselines, and ablation studies. Consequently, no test song appears in either the SFT data or the reward-learning data.

Offline critique generation. The original SongEval examples contain only song audio and five-dimensional mean expert ratings. To construct $\mathcal{D}_{\text{off}}$, we provide Gemini-3-Pro with the song, definitions of the five dimensions, and the expert mean ratings. The prompt asks the model to infer perceptible musical evidence consistent with the scores on a five-point Likert scale, including melodic repetition and variation, section-level energy, rhythmic density, harmonic progression, vocal breathing, and changes in arrangement. The critique must neither state numerical scores nor invent the identity of a singer or band. We retain the generated aesthetic critique together with the original expert ratings.

Human verification of LLM-generated critiques. After critique generation and before SFT, a panel of eight music experts manually reviews the critiques generated by Gemini-3-Pro to determine whether they are consistent with human perceptual judgments of the corresponding songs. For each critique, the reviewers listen to the corresponding song and assess whether the commentary is grounded in audible musical evidence. Factual accuracy and the absence of unsupported or hallucinated claims serve as the primary acceptance criteria.

The review additionally considers the depth of the analysis, whether its claims refer to perceptible musical phenomena, whether all five evaluation dimensions are adequately covered, and whether the critique is logically coherent and clearly written. Only critiques accepted through this verification process are retained as SFT data in $\mathcal{D}_{\text{off}}$.

Self-generated critique construction. After the first SFT epoch, we run the resulting MOSS-Audio checkpoint over all 2,199 training songs. We use `do_sample=False`, `num_beams=1`, and at most 4,096 new tokens, corresponding to greedy decoding with $T = 0$. We replace each teacher critique with the corresponding self-generated critique while preserving the audio path, evaluation prompt, and expert mean ratings, thereby obtaining $\mathcal{D}_{\text{on}}$.

B. Gemini Model Versions

The external teacher referred to as Gemini-3-Pro in the main text uses the model identifier `gemini-3-pro-preview`. The in-domain LLM-as-a-Judge baseline referred to as Gemini-3.1-Pro uses `gemini-3.1-pro-preview`.

C. Complete Training Configuration

Critique supervised fine-tuning. We fully fine-tune MOSS-Audio-8B-Instruct on $\mathcal{D}_{\text{off}}$ using four NVIDIA H200 GPUs, bfloat16 precision, a per-device batch size of 1, and gradient accumulation over

8 steps, giving a global batch size of 32. The learning rate is 5×10^{-5} with no weight decay, a 5% warmup, cosine scheduling, a maximum sequence length of 10,000, and random seed 42. Although the launch configuration specifies five epochs, we use checkpoint-69, saved after the first epoch, to initialize both the critic and the reward backbone; we therefore report one effective SFT epoch.

MuseCritic reward learning. Starting from SFT checkpoint-69, we train MuseCritic for ten epochs on $\mathcal{D}_{\text{on}}$ using four NVIDIA H200 GPUs and bfloat16 precision. The backbone is updated with LoRA of rank 8 and scaling factor 32, while the reward head is fully trainable with dropout probability 0.1. We use a learning rate of 2×10^{-4} , weight decay of 0.1, a per-device batch size of 1, gradient accumulation over 8 steps, a global batch size of 32, a 5% warmup, cosine scheduling, a maximum sequence length of 10,000, and random seed 42.

SongEval baseline training. The in-domain SongEval (UTMOS-based) baseline is fully trained for ten epochs on two NVIDIA H200 GPUs using bfloat16 precision, a learning rate of 1×10^{-4} , weight decay of 0.1, a per-device batch size of 1, gradient accumulation over 8 steps, and a global batch size of 16. It uses the same 5% warmup, cosine scheduling, and random seed 42.

Reward-head implementation. The final valid-token state from the last backbone layer first passes through dropout with probability 0.1, followed by a linear projection from $\mathbb{R}^d$ to $\mathbb{R}^5$ and a sigmoid range transformation that produces scores in $(1, 5)^5$. Reward learning uses DeepSpeed ZeRO-3. LoRA is applied to the language backbone, while the reward head remains fully trainable. Training uses only the MSE objective in Equation 10, without an auxiliary critique language-modeling loss.

Checkpoints and inference. We save a checkpoint after every reward-learning epoch and use checkpoint-690 from the end of epoch ten for the main results. During in-domain testing, the model generates at most 4,096 new tokens using greedy decoding with $T = 0$. The five rewards are computed from the hidden state of the final valid token—the end-of-turn token following the critique—in the complete song–rubric–self-generated-critique sequence.

D. Aesthetic-Evaluation Prompt Templates

Figure 3 presents the prompt used to construct $\mathcal{D}_{\text{off}}$. At request time, the five variables in braces are replaced with the corresponding expert mean scores for the song, and the song audio is attached to the request.

Figure 4 presents the user prompt used for both the $\mathcal{D}_{\text{on}}$ and $\mathcal{D}_{\text{off}}$ training examples. This prompt is used for MuseCritic critique SFT, online critique generation, and subsequent reward modeling. The `<audio>` marker indicates the position at which the model processor inserts the song audio.

E. Music Arena Evaluation Details

Each Music Arena example contains a chosen and rejected song generated under the same text condition. For MuseCritic and SongEval with the official MuQ-based weights, we obtain five-dimensional scores for each song and take their arithmetic mean. For Audiobox Aesthetics, we average its four aesthetic dimensions. For Qwen3-Omni, we parse and average its five generated ratings. An example is correct only if $r(x^+) > r(x^-)$; ties count as errors. Final accuracy is the fraction of correct predictions across all 733 preference pairs.

You are an expert music critic.

Listen to a song and write a professional, detailed critique grounded in audible musical phenomena while taking the human experts' ratings as given. Return the response in Simplified Chinese.

The five evaluation dimensions are:

1. Overall Coherence
Evaluates musical and emotional continuity across the intro, verses, choruses, and outro. High ratings reflect smooth transitions, consistent dynamics, and a unified emotional tone throughout the piece.
2. Memorability
Evaluates distinctive features—such as a catchy melody, rhythmic motif, or lyrical hook—that make the song easy to remember after one listen.
3. Naturalness of Vocal Breathing and Phrasing
Evaluates phrasing quality and breath control, including alignment with semantic boundaries and rhythmic cues and whether breathing supports a fluent delivery without disrupting the singing flow.
4. Clarity of Song Structure
Evaluates whether recognizable sections (e.g., verse, chorus, bridge) are clearly delineated and logically organized. Conventional and novel forms may both rate highly when their segmentation is musically meaningful.
5. Overall Musicality
Evaluates listening enjoyment based on melody, harmony, instrumentation, and the integration of vocals and accompaniment.

Requirements:

1. Use professional terminology from music theory and audio production.
2. Analyze each dimension separately. Begin each analysis with its number and name, e.g., "1. Overall Coherence: ".
3. Ground each analysis in concrete musical evidence consistent with the given rating, such as melodic repetition/variation, section-level energy, rhythmic density, harmonic progression, vocal breaths, or arrangement.
4. Do not reassess the ratings. Explain them through professional, specific, and interpretable comments based on audible musical phenomena.
5. Do not state numerical ratings in the critique or say that a phenomenon occurs "because the rating is high/low."
6. Discuss only the song. Do not invent the singer's or band's identity.
7. The expert team's ratings (1–5) are:
 - Coherence: {coherence}
 - Memorability: {memorability}
 - Naturalness: {naturalness}
 - Clarity: {clarity}
 - Musicality: {musicality}
 After the critique, reproduce these ratings in a JSON code block. The scores are means across annotators and may therefore be non-integer values.
8. Ratings express agreement on a five-point Likert scale. Infer musical evidence according to the following mapping:
 - 5 (Excellent): outstanding quality with no discernible shortcomings.
 - 4 (Good): strong quality with only minor room for improvement.
 - 3 (Fair): adequate but unremarkable baseline quality.
 - 2 (Poor): clear shortcomings that impair the overall effect.
 - 1 (Very Poor): severe problems that fail to meet the criterion.

Output format:

Return valid JSON following this format, without an introduction or summary.

```
{
  "critical": "Evaluate the song along all five dimensions...",
  "scores": {
    "coherence": {coherence},
    "memorability": {memorability},
    "naturalness": {naturalness},
    "clarity": {clarity},
    "musicality": {musicality}
  }
}
```

Now evaluate the song.

Figure 3 | Prompt used by Gemini-3-Pro to generate offline aesthetic critiques. At request time, the five variables in braces are replaced with the corresponding expert mean scores, and the audio is supplied concurrently. The expert scores constrain the evaluative polarity of the critique for each dimension, while the prompt explicitly prohibits numerical ratings in the critique text.

You are an expert music critic.

Listen to a song and write a professional, detailed critique in Simplified Chinese, grounded in audible musical phenomena.

The five evaluation dimensions are:

1. Overall Coherence
Evaluates musical and emotional continuity across the intro, verses, choruses, and outro. High ratings reflect smooth transitions, consistent dynamics, and a unified emotional tone throughout the piece.
2. Memorability
Evaluates distinctive features—such as a catchy melody, rhythmic motif, or lyrical hook—that make the song easy to remember after one listen.
3. Naturalness of Vocal Breathing and Phrasing
Evaluates phrasing quality and breath control, including alignment with semantic boundaries and rhythmic cues and whether breathing supports a fluent delivery without disrupting the singing flow.
4. Clarity of Song Structure
Evaluates whether recognizable sections (e.g., verse, chorus, bridge) are clearly delineated and logically organized. Conventional and novel forms may both rate highly when their segmentation is musically meaningful.
5. Overall Musicality
Evaluates listening enjoyment based on melody, harmony, instrumentation, and the integration of vocals and accompaniment.

Requirements:

1. Use professional terminology from music theory and audio production.
2. Analyze each dimension separately. Begin each analysis with its number and name, e.g., "1. Overall Coherence: ".
3. Ground each dimension in concrete musical evidence, such as melodic repetition/variation, section-level energy, rhythmic density, harmonic progression, vocal breaths, or changes in arrangement.
4. Provide professional, specific, and interpretable comments based on audible musical phenomena.
5. Discuss only the song. Do not invent the singer's or band's identity.

Now evaluate the song.
<audio>

Figure 4 | Prompt used to train the SFT-Critique-Generator, generate online aesthetic critiques with MOSS-Audio, and perform MuseCritic reward-model training. The model generates critiques along five dimensions without receiving expert scores; <audio> indicates the audio input position.

Window	Model	Audiobox Aesthetics ↑				SongEval ↑				
		CE	CU	PC	PQ	CO	MU	ME	CL	NA
Full audio	Muse	6.79	7.50	5.77	7.93	3.76	3.56	3.72	3.65	3.62
	MUSE-GRPO	7.03	7.55	6.05	7.98	3.89	3.68	3.84	3.77	3.73
360 s	Muse	6.93	7.53	5.92	7.97	3.85	3.65	3.81	3.73	3.70
	MUSE-GRPO	7.10	7.56	6.13	8.00	3.94	3.73	3.89	3.82	3.77
330 s	Muse	7.00	7.55	6.00	7.99	3.88	3.69	3.83	3.76	3.72
	MUSE-GRPO	7.16	7.58	6.19	8.02	3.96	3.76	3.91	3.84	3.79
300 s	Muse	7.10	7.58	6.11	8.03	3.92	3.73	3.87	3.80	3.76
	MUSE-GRPO	7.24	7.60	6.29	8.04	3.99	3.80	3.94	3.87	3.82
270 s	Muse	7.21	7.60	6.24	8.06	3.96	3.78	3.90	3.84	3.79
	MUSE-GRPO	7.34	7.63	6.40	8.08	4.03	3.84	3.97	3.91	3.85
240 s	Muse	7.33	7.64	6.37	8.11	4.01	3.84	3.94	3.89	3.83
	MUSE-GRPO	7.45	7.66	6.52	8.12	4.07	3.89	4.00	3.94	3.89

Table 5 | **Robustness of downstream aesthetic evaluation across audio windows.** Both models are evaluated on the same 100 Muse test prompts. “Full audio” uses each complete generation; the remaining rows apply a common maximum-duration prefix, leaving shorter songs unchanged. Metric abbreviations follow Table 3. Bold marks the higher observed value within each window.

F. Muse-GRPO Training and Evaluation Details

Training data. We sample 500 multi-turn song-generation examples from the public Muse training data with seed 42 while preserving the original language distribution: 305 English and 195 Chinese examples. All inputs retain the Muse multi-turn prompt format, and the policy is initialized from Muse-0.6B.

GRPO configuration. Training uses six NVIDIA H200 GPUs, a per-device batch size of 2, gradient accumulation over 2 steps, eight candidate generations, a learning rate of 10^{-6} , a 5% warmup, and one epoch. The vLLM rollout server uses two NVIDIA H200 GPUs, a maximum model length of 20,000 and GPU memory utilization of 0.7. Sampling uses temperature 0.9, top- p 0.9, and a repetition penalty of 1.3. Each completion contains at most 3,000 tokens, and a complete multi-turn trajectory contains at most 20,000 tokens. The reward for each candidate song is the mean of the five MuseCritic scores; within-group relative advantages are computed according to GRPO.

Evaluation protocol. We evaluate on the 100 multi-turn test prompts from Muse (Jiang et al., 2026) and apply identical greedy decoding with $T = 0$ to both the original Muse and the trained MUSE-GRPO. Under modality-extended generation, deterministic decoding can exhibit inference-time instability on some evaluation examples: the model may fail to produce a valid ending and instead enter token-level repetition, producing an extended noisy tail. This behavior does not consistently occur on a fixed subset of the evaluation prompts. To prevent this decoding artifact from dominating the aesthetic assessment of otherwise valid song content while retaining all 100 test examples, we adopt a fixed-prefix protocol. All generations longer than 240 seconds are evaluated using their first 240 seconds, whereas shorter generations remain unchanged and are neither padded nor looped. The same deterministic preprocessing is applied to Muse and MUSE-GRPO. We then compute the nine metrics in Table 3 with Audiobox Aesthetics and SongEval, focusing the main comparison on the aesthetic quality of the valid musical content.

Robustness across evaluation windows. As a robustness check, we evaluate both complete, untruncated generations and fixed-prefix windows of 240, 270, 300, 330, and 360 seconds. Table 5 shows that MUSE-GRPO obtains higher observed values than Muse on all nine metrics at every evaluation window, including the complete audio. The 240-second protocol reduces the observed Muse–MUSE-GRPO gap on every metric relative to the untruncated evaluation, indicating that the main results do not depend on excluding the noisy tails and provide a conservative estimate of the relative improvement.

G. Reproducibility Statement

We use random seed 42 for data splitting, model training, and sampling. Online critique generation, in-domain testing, Music Arena comparisons, and song generation for the 100 downstream prompts all use deterministic decoding to eliminate sampling variance in model comparisons. The fixed-prefix protocol described in Appendix F is used to handle occasional degenerate tails in downstream generations.

H. Ethics Statement

This study follows applicable data-privacy regulations and ethical standards in data acquisition and experimental evaluation. Both the training and evaluation sets are constructed from public datasets,

and the study uses no private personal data. Given the data types and scope of this work, we identify no additional substantial ethical risks.

I. AI Assistants in Research or Writing

During the preparation of this manuscript, AI assistive technologies were utilized exclusively to enhance linguistic clarity, readability, and stylistic presentation. Specifically, these tools were employed to polish phrasing and refine the text. AI played no role in the research conceptualization, methodology design, data collection, data analysis, interpretation of results, or the formulation of scientific conclusions. The authors assume full responsibility for all intellectual contributions and the final content of this work.